UPDATE YOUR PRIORS · THE STORY OF AN IDEA

Every Model Begins with a Prior

How Bayes's theorem was buried, rescued, banned and enthroned — and came back to run the age of AI.

1761 · Buried 1763 · Rescued 1774 · Laplace 1800s · Vanishes 1925 · Fisher's Empire Today · The Return

A presentation companion to the essay of the same name



ACT I · THE DISCOVERY — TUNBRIDGE WELLS, 1761

The letter he never sent

A dissenting minister sits down to write one of the strangest letters in history. He writes it to himself. He never sends it, never publishes it.

When Thomas Bayes dies in 1761, the essay is found among his personal papers — polished, worked, complete. Tucked away.

1761

the theorem is written —
and filed away

2 yrs

later, he is dead —
the world still hasn't seen it

260+

years later, it runs
your inbox, your GPS, your AI

He solved the deepest problem in reasoning. Then he put it in a drawer.

Life is lived backwards: evidence first, causes second

You wake up in a room with no windows. You know nothing about the weather outside.

Someone walks in **dripping wet**.

Instantly — before any calculation — you have reasoned from an observation to a revised belief about the world. Every mind does this. No one could say *how much* the wet coat should change your mind.



rain? maybe 50 : 50 the dripping coat

belief: rain — much more likely

...but precisely how much more?

Bayes's question: put an exact number on the update.

Inverse probability — the mathematics that didn't exist

🩺 The doctor

sees a symptom and reasons backwards to a diagnosis.

🪨 The geologist

sees a rock formation and reasons backwards to a history of the Earth.

🔭 The astronomer

sees light from a distant star and reasons backwards to what the star is made of.

Science works forwards: model → prediction → test. But practice runs backwards — and for centuries there was no rigorous mathematics for it. Philosophers argued. Mathematicians threw up their hands.

Until a minister in Tunbridge Wells quietly solved it — as a game.



ACT I · THE METHOD

The billiard table: learning about what you cannot see

1. Your back is turned.
2. A first ball lands — unseen.
3. Each new ball lands, and all you are told is whether it fell to the *left* or the *right* of the first.
4. The golden band is your uncertainty — watch what evidence does to it.

Every judgment works like this: prior → evidence → posterior.

One equation for changing your mind

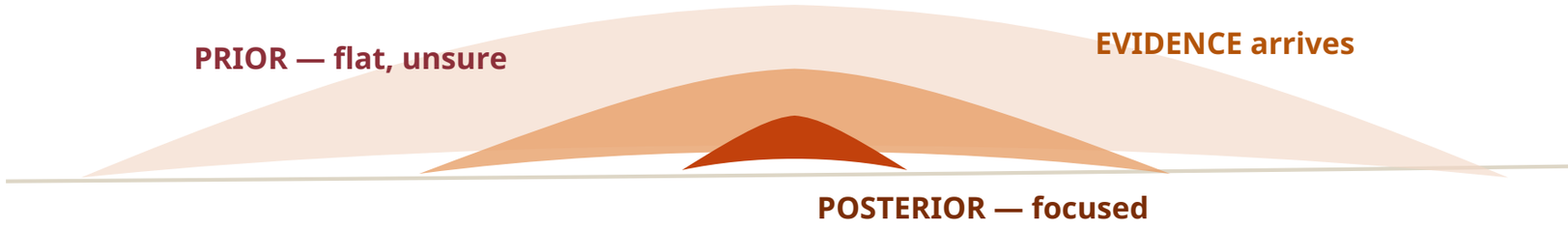
$$P(H \mid E) = \frac{P(E \mid H) \times P(H)}{P(E)}$$

POSTERIOR — your new belief

LIKELIHOOD — the evidence, if you're right

PRIOR — what you believed before

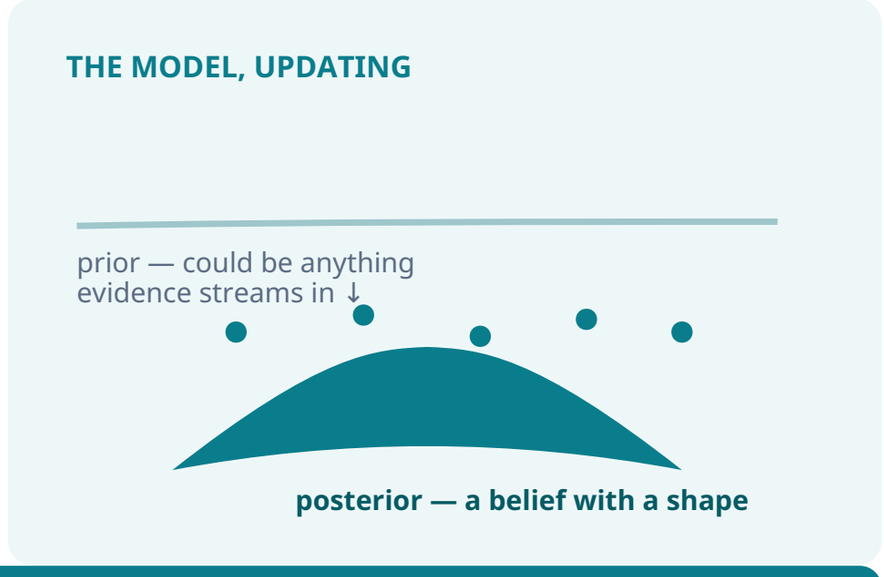
EVIDENCE — the weight of what you saw



New belief = old belief, updated by exactly what the evidence is worth.

Every model begins with a prior

1. An AI model begins as a **prior**: an expectation about the world, built from everything it trained on.
2. Then evidence arrives — new data, a correction, a reward — and the original beliefs shift, weighted by surprise: the more the surprise, the more it shifts.
3. **Training** is updating at industrial scale. **Fine-tuning** is a narrower, stronger stream of evidence.
4. The best example is the **spam filter** — the purest Bayes of all: a prior on what is junk mail, then word by word it declares whether it's junk or not.



The billiard table is how minds work — biological and artificial alike.

ACT I · THE PUZZLE

Why did Bayes bury a result like this?

Reason one — his doubt

Where does the prior come from? On a billiard table, "equally likely" is defensible. In the real world, pick the wrong starting number and the theorem can mislead you arbitrarily.

He may have refused to publish what he could not fully defend.

Reason two — the danger

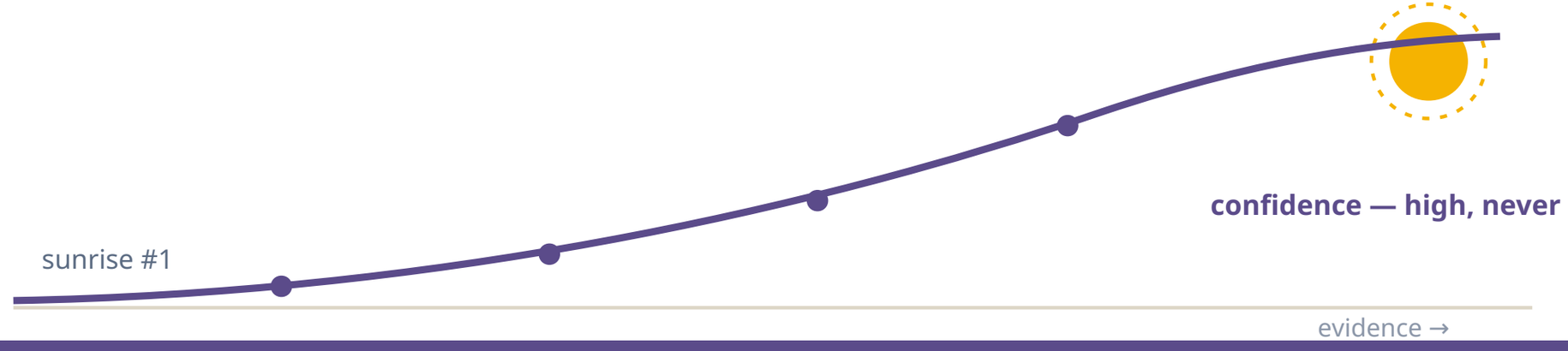
Probability was tangled with **theology**. Hume had argued no testimony could overcome the prior improbability of a miracle.

Bayes's theorem could be weaponised in that fight — by either side. A quiet minister may have wanted no part of it.

The thing he doubted was the heart of his own theorem: where the first belief comes from.

Richard Price — the friend who found Bayes's manuscript

Richard Price — minister, radical, defender of revolutions — found the manuscript, mastered it, expanded it, and read it to the Royal Society in 1763. His addition was the deepest claim of all: **induction**. The sun has risen every day of our lives. Each sunrise is a data point. Watch confidence grow:

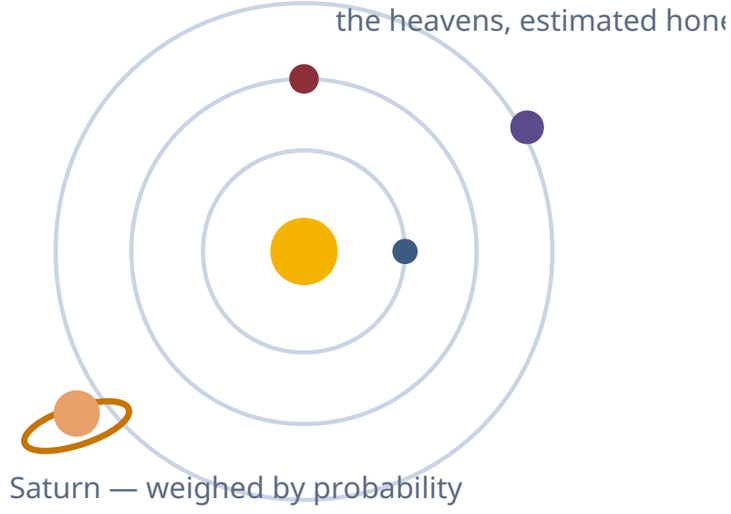


Not to certainty — never to certainty — but precisely as confident as the evidence warrants.

Laplace: the man who built the system

A generation later, Laplace finds the same theorem **independently** — and builds it into a system for reasoning about everything: the mass of Saturn from noisy observations, the ratio of boys to girls born in Paris, the probability the sun rises tomorrow.

Napoleon: why does your book on the heavens never mention God?
"I had no need of that hypothesis."



Where Bayes hesitated, Laplace built an empire of evidence.

"Assume everything equally likely" is still a choice

- Laplace's answer to the prior problem: the **principle of indifference**
- No information? Split the probability evenly
- A starting point — not an escape from choosing
- AI makes the same move today, usually without admitting it:

The architecture

is a prior about what patterns exist.

The training data

is a prior about what the world contains.

The fine-tuning

is a prior about what matters.

The prior is the accumulated knowledge of the system, made explicit. State it — or smuggle it in.

Prior belief: banned from science, never disproved

The nineteenth century falls in love with **positivism**: only what can be directly observed and measured belongs in science. The prior is unobservable, personal, different in different hands — *metaphysics*. And metaphysics is banned.



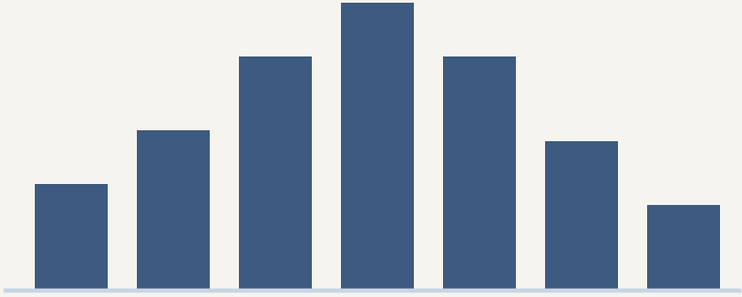
The idea was never disproved. It was ruled out of bounds — and it faded.

Pearson rebuilds probability on frequency alone

Karl Pearson — founder of *Biometrika*, pope of British statistics — redefines probability as **long-run frequency**: how often things happen in repeated trials. Clear. Testable. Reproducible.

And single cases? Hypotheses? Beliefs? Frequencies don't apply — so they're defined out of science.

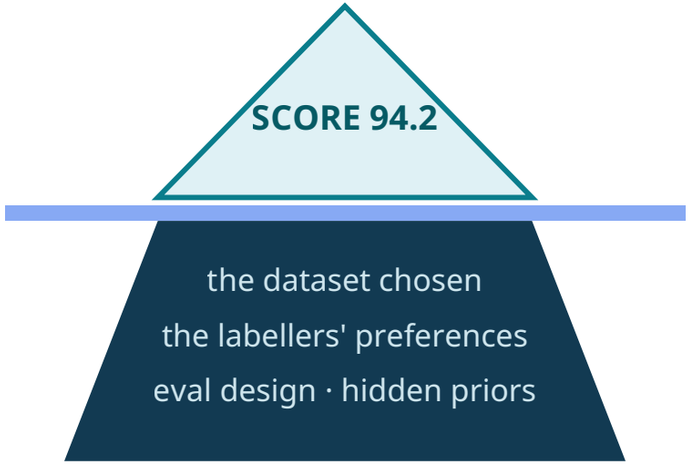
10,000 trials — the curve is real



but what about the single case in front of you?

The priors didn't disappear. They went underground — into the methodology, where no one could question them.

The new positivism: if it isn't on the benchmark, it isn't real



When only benchmark scores count as real, everything a benchmark can't measure — judgement, taste, robustness in the strange case, the long tail where the world actually lives — is quietly treated as unreal.

And the beliefs don't vanish. They sink below the waterline: into the dataset, the evaluation, the preferences of the people labelling the examples.

Invisible beliefs are harder to criticise than visible ones. True in 1901. True in every model card today.



ACT IV · THE EMPIRE — ROTHAMSTED, 1925

Fisher builds an empire in a field of wheat

Plots of wheat, growing at wildly different rates. Does the fertiliser work — or is it luck? Fisher's answer: **design first, randomise, decide in advance what counts as evidence.** It became the foundation of every randomised trial since.

Brilliant machinery — built to keep belief out of the result.

The teacup test

A lady at Rothamsted claims she can taste whether the milk or the tea was poured first. Fisher tests her: **8 cups, 4 of each kind**.

Ways to choose 4 from 8: C(8,4) = **70** — so a perfect score by pure guessing is **1 in 70 ≈ 1.4%**, one golden ticket among seventy. **She got all eight right.**



70 possible choices — **1** is a perfect score



But Fisher never asks the prior question: how likely was her claim before the test?

Significance is not plausibility

Fisher's question

"Would this result happen by chance, *if there were no effect?*"

The p-value. It tells you about **chance**.

Bayes's question

"Given everything I knew before — *how likely is the claim to be true?*"

It tells you about **plausibility**.



The gate at **0.05** — a convention, not a law of nature. An effect can be significant and trivial; large and "not significant".

"Is it significant?" quietly replaced "Is it real — and does it matter?"

⚡ WHAT IT MEANS FOR AI · 3 OF 3

Living in Fisher's empire

AI judges itself with Fisher's tools: A/B tests, significance thresholds, leaderboards where a fraction of a point crowns the better model. And it falls into Fisher's trap daily — gains that are significant and trivial, results that evaporate under a different test.

The Bayesian question is the one the field keeps needing: given the base rates, the plausibility, the track record — **how much should this evidence actually move us?**

THE HEADLINE CLAIM

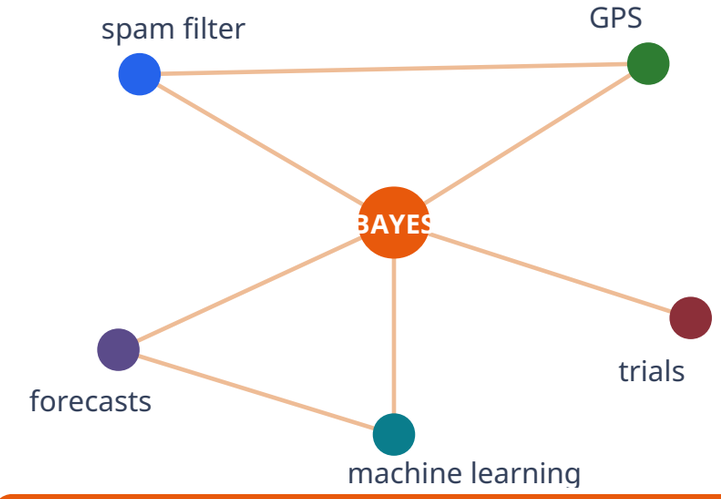
"New model beats the benchmark by 0.3 points!!"

THE BAYESIAN REPLY

Low prior? Then I need overwhelming evidence — not a p-value under a line.

State your prior. Meet the evidence. Update by exactly what it's worth. That is the operating manual for the age of AI.

The ending Bayes never saw: his idea won anyway



Bayesian reasoning now runs the spam filter and the GPS, the clinical trial and the weather forecast, the diagnostic system and the recommender — the probabilistic machinery underneath modern machine learning itself.

Fisher's empire also remains: statistics is still taught his way, and "significant" is still misread as "probably true".

Every learning system converges on learning from experience. We need both questions — and to know which is which.



THE LESSON OF THE DRAWER

State your prior.

Meet the evidence — all of it, especially what surprises you.

Update by exactly as much as the evidence warrants.

The prior is not the bug in the system. It is the feature — how a doctor, a scientist, a model, a mind, becomes less wrong over time.

The hardest and most Bayesian thing a mind can do: change it, on purpose, by the right amount.